FIITJEE

ALL INDIA INTEGRATED TEST SERIES

HALF COURSE TEST – I

JEE (Main)-2020

TEST DATE: 30-12-2018

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The unit of Stefan-Boltzmann's constant (σ) is:

(A) $\frac{watt^4}{m \times K^4}$

(B) $\frac{calorie}{m^2 \times K^4}$

(C) $\frac{watt}{m^2 \times K^4}$

(D) $\frac{joule}{m^2 \times K^4}$

2. The equation of the stationary wave is:

$y = 2A \sin\left(\frac{2\pi ct}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$.

Which of the following statements is wrong?

(A) The unit of ct is same as that of λ

(B) The unit of x is same as that of λ

(C) The unit of $2\pi c / \lambda$ is same as that of $2\pi x / \lambda t$

(D) The unit of c / λ is same as that of x / λ

3. Given that: $y = a \cos\left(\frac{t}{p} - qx\right)$, where t represents time in second and x represents distance in metre. Which of the following statements is true?

(A) The unit of x is same as that of q

(B) The unit of x is same as that of p

(C) The unit of t is same as that of q

(D) The unit of t is same as that of p

4. A spherical body of mass m and radius r is allowed to fall in a medium of viscosity η. The time in which the velocity of the body increases from zero to 0.63 times, the terminal velocity (v) is called time constant (τ). Dimensionally τ can be represented by:

(A) $\frac{mr^2}{6\pi\eta}$

(B) $\sqrt{\left(\frac{6\pi mr\eta}{g^2}\right)}$

(C) $\frac{m}{6\pi\eta rv}$

(D) none of these

5. The pressure on a square plate is measured by measuring the force on the plate and the length of the sides of plate. If the maximum error in the measurement of force and length are respectively 4% and 2%, the maximum error in the measurement of pressure is:

(A) 1%
(B) 2%
(C) 6%
(D) 8%

6. If $Q = \frac{X^n}{Y^m}$ *and* ΔX is absolute error in the measurement of X, ΔY is absolute error in the measurement of Y, then absolute error ΔQ *in* Q is:

(A) $\Delta Q = \pm\left(n\frac{\Delta X}{X} + m\frac{\Delta Y}{Y}\right)$

(B) $\Delta Q = \pm\left(n\frac{\Delta X}{X} + m\frac{\Delta Y}{Y}\right)Q$

(C) $\Delta Q = \pm\left(n\frac{\Delta X}{X} - m\frac{\Delta Y}{Y}\right)Q$

(D) $\Delta Q = \pm\left(n\frac{\Delta X}{Y} - m\frac{\Delta Y}{X}\right)Q$

7. A particle of mass $= 5\ units$ is moving with a uniform speed $V = 3\sqrt{2}\ units$ in the XOY Plane along the line $Y = X + 4$. The magnitude of the angular momentum of the particle about the origin is:

(A) 60 unit
(B) $40\sqrt{2}\ unit$
(C) zero
(D) $7.5\ unit$

8. A particle of mass m is projected with a velocity V making an angle of 45° with the horizontal. The magnitude of the angular momentum of the projectile about the point of projection when the particle is at its maximum height h is:

(A) zero
(B) $\frac{mV^2}{\sqrt{2}\ g}$
(C) $\frac{mV^2}{4\sqrt{2}g}$
(D) $m\sqrt{2gh^3}$

9. The displacement of a particle after time t is given by $x = (k/b^2)(1-e^{-bt})$, *where* b is a constant. What is the acceleration of the particle?

(A) ke^{-bt}

(B) $-ke^{-bt}$

(C) $\frac{k}{b^2}e^{-bt}$

(D) $\frac{-k}{b^2}e^{-bt}$

10. The deceleration experienced by a moving motor boat, after its engine is cut-off is given by: $\frac{dv}{dt} = -kv^3$, *where* k is constant. If v_0 is the magnitude of the velocity at cut-off, the magnitude of the velocity at a time t after the cut-off is:

(A) $v_0 / 2$

(B) v_0

(C) $v_0 e^{-k\,t}$

(D) $\frac{v_0}{\sqrt{(2v_0^2 kt + 1)}}$

11. The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will:

(A) be independent of β

(B) drop to zero when $\alpha = \beta$

(C) go on decreasing with time

(D) go on increasing with time

12. The equation of motion of a projectile is:

$y = 12x - \frac{5}{9}x^2$.

The horizontal component of velocity is $3\ ms^{-1}$. Given that $g = 10\ ms^{-2}$, what is the range of the projectile?

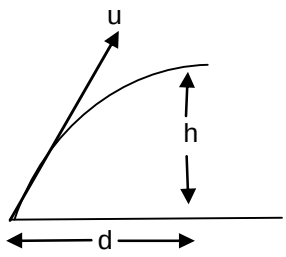
(A) 12.4 m

(B) 21.6 m

(C) 30.6 m

(D) 36.0 m

13. If a stone is to hit at a point which is at a distance d away and at a height h above the point from where the stone starts, then what is the value of initial speed u if the stone is launched at an angle θ?



(A) $\frac{g}{\cos\theta}\sqrt{\frac{d}{2(d\ \tan\theta - h)}}$

(B) $\frac{d}{\cos\theta}\sqrt{\frac{g}{2(d\ \tan\theta - h)}}$

(C) $\sqrt{\frac{gd^2}{h\cos^2\theta}}$

(D) $\sqrt{\frac{gd^2}{(d-h)}}$

14. The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times its initial speed. If the range of the projectile is P times the maximum height attained by it, P is equal to:

(A) $\frac{4}{3}$

(B) $2\sqrt{3}$

(C) $4\sqrt{3}$

(D) $\frac{3}{4}$

15. The potential energy for a conservative system is given by:
$U = ax^2 - bx.$
Where a and b are positive constants. The law of the force governing the system is:

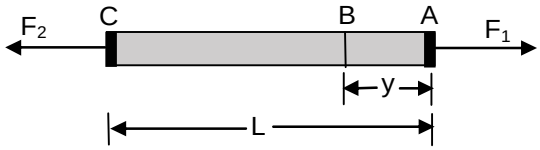
(A) $F = constt.$

(B) $F = bx - 2a$

(C) $F = b - 2ax$

(D) $F = 2ax$

16. A rod of length L and mass M is acted on by two unequal forces F_1 *and* F_2 $(< F_1)$ as shown in the following figure.



The tension in the rod at a distance y from the end A is given by:

(A) $F_1\left(1-\frac{y}{L}\right)+F_2\left(\frac{y}{L}\right)$

(B) $F_2\left(1-\frac{y}{L}\right)+F_1\left(\frac{y}{L}\right)$

(C) $(F_1-F_2)\frac{y}{L}$

(D) none of these

17. A force acts on a 3.0 g particle in such a way that the position of the particle as a function of time is given by:

$x = 3t - 4t^2 + t^3.$

Where x is in metres and t is in seconds. The work done during the first 4 s is:

(A) 570 mJ
(B) 450 mJ
(C) 490 mJ
(D) 528 mJ

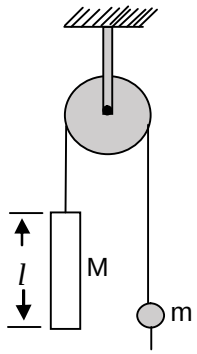
18. A bullet is fired from a gun. The force on the bullet is given by:

$F = 600 - 2 \times 10^5 \ t.$

Where F is in newton and t in second. The force on the bullet becomes zero as soon as it leaves the barrel. What is the average impulse imparted to the bullet?

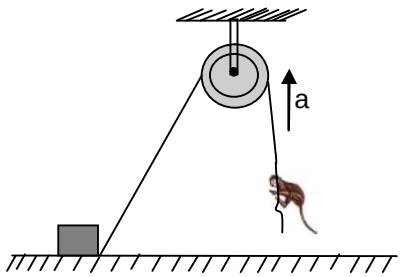
(A) $9 \ N - s$
(B) $58.8 \ kg / s$
(C) $0.9 \ kg / s$
(D) $178.4 \ kg / s$

19. Figure shows an arrangement of a rod of length $l \ and \ mass \ M$ and a bead of mass m attached to a weightless string passing over a frictionless pulley. At $t = 0$, the bead is in level with the lower end of the rod. The bead slides down the string with considerable friction and is opposite to the other end of the rod after T second. Assuming friction between the bead and the string to be constant all through, the frictional force is:



(A) $\dfrac{2Mm}{(M-m)} \dfrac{l}{t^2}$
(B) $\dfrac{(M-m)}{2Mm} \dfrac{l}{t^2}$
(C) $\dfrac{2Mm}{(M+m)} \dfrac{l}{t^2}$
(D) $\dfrac{(M+m)}{2Mm} \dfrac{l}{t^2}$

20. A light string fixed at one end to a clamp on ground passes over a fixed pulley and hangs at the other side. It makes an angle of 30° with the ground. A monkey of mass 5 kg climbs up the rope. The clamp can tolerate a vertical force of 40 N only. The maximum acceleration in upward direction with which the monkey can climb safely is: (neglect friction and take $g = 10 m / s^2$)



(A) $2 \ m / s^2$
(B) $4 \ m / s^2$
(C) $6 \ m / s^2$
(D) $8 \ m / s^2$

21. An elevator accelerates upwards at a constant rate. A uniform string of length L and mass m supports a small block of mass M that hangs from the ceiling of the elevator. The tension at distance l from the ceiling is T. The acceleration of the elevator is:

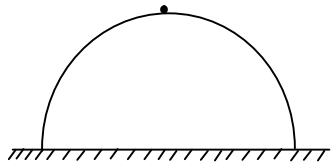
(A) $\dfrac{T}{M+m-\dfrac{ml}{L}}-g$

(B) $\dfrac{T}{2M+m-\dfrac{ml}{L}}+g$

(C) $\dfrac{T}{M+\dfrac{ml}{L}}-g$

(D) $\dfrac{T}{2M-m+\dfrac{ml}{L}}-g$

22. A hemisphere of radius *R* and of mass $4m$ is free to slide with its base on a smooth horizontal table. A particle of mass m is placed on the top of the hemisphere. The angular velocity of the particle relative to hemisphere at an angular displacement θ when velocity of hemisphere v is:



(A) $\dfrac{5v}{R\cos\theta}$

(B) $\dfrac{2v}{R\cos\theta}$

(C) $\dfrac{3v}{R\sin\theta}$

(D) $\dfrac{5v}{R\sin\theta}$

23. Three particles each of mass M are placed at the three corners of an equilateral triangle of side l. What is the force due to this system of particles on another particle of mass m placed at the mid -point of any side?

(A) $\dfrac{3GMm}{4l^2}$

(B) $\dfrac{4GMm}{3l^2}$

(C) $\dfrac{GMm}{4l^2}$

(D) $\dfrac{4GMm}{l^2}$

24. The two planets have radii r_1 *and* r_2 and their densities p_1 *and* p_2 respectively. The ratio of acceleration due to gravity on them will be:

(A) $r_1 p_1 : r_2 p_2$

(B) $r_1 p_1^2 : r_2 p_2^2$

(C) $r_1^2 p_1 : r_2^2 p_2$

(D) $r_1 p_2 : r_2 p_1$

25. If T is the surface tension of a liquid, the energy needed to break a liquid drop of radius R into 64 drops, is:

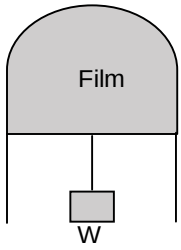
(A) $6\pi R^2 T$

(B) $\pi R^2 T$

(C) $12\pi R^2 T$

(D) $8\pi R^2 T$

26. A thin liquid film formed between a U-shaped wire and a light slider supports a weight of $1.5\times10^{-2}\ N$ (see figure). The length of the slider is 30 cm and its weight are negligible. The surface tension of the liquid film is:



(A) $0.0125\ Nm^{-1}$

(B) $0.1\ Nm^{-1}$

(C) $0.05\ Nm^{-1}$

(D) $0.025\ Nm^{-1}$

27. A small metal sphere of radius a is falling with a velocity v through a vertical column of a viscous liquid. If the coefficient of viscosity of the liquid is η, then the sphere encounters an opposing force of:

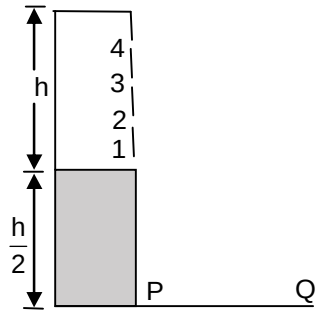
(A) $6\pi\eta a^2 v$

(B) $\dfrac{6\eta v}{\pi a}$

(C) $6\pi\eta a v$

(D) $\dfrac{\pi\eta v}{6a^3}$

28. A cylinder of height h is filled with water and is kept on a block of height $h/2$. The level of water in the cylinder is kept constant. Four holes numbered 1, 2, 3 and 4 are at the side of the cylinder and at heights 0, $h/4$, $h/2$ *and* $3h/4$ respectively. When all four holes are opened together, the hole from which water will reach farthest distance on the plane PQ is the hole number:



(A) 1
(B) 2
(C) 3
(D) 4

29. Assuming the sun to be a spherical body of radius R at a temperature of TK, evaluate the total radiant power, incident on the earth, at a distance r from the sun, is:
(Where r_0 is the radius of the earth and σ is Stefan's constant.)

(A) $\frac{R^2\sigma T^4}{r^2}$

(B) $\frac{4\pi r_0^2 R^2\sigma T^4}{r^2}$

(C) $\frac{\pi r_0^2 R^2\sigma T^4}{r^2}$

(D) $\frac{r_0^2 R^2\sigma T^4}{4\pi r^2}$

30. The length of an open organ pipe is twice the length of another closed organ pipe. The fundamental frequency of the open pipe is 100 Hz. The frequency of the third harmonic of the closed pipe is:
(A) 100 Hz
(B) 200 Hz
(C) 300 Hz
(D) 150 Hz

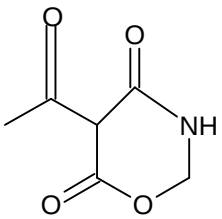
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

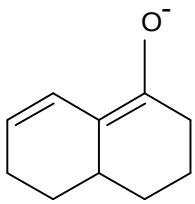
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

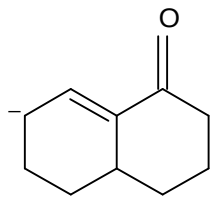
31. Identify which functional group is not present in following compound?

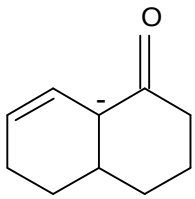


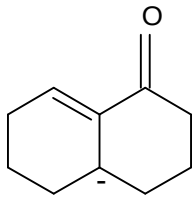
(A) Ketone
(B) Ester
(C) Amide
(D) Ether

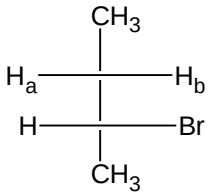
32. Which of the following is not a resonating structure of others?

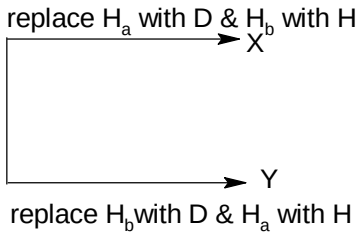
(A) 

(B) 

(C) 

(D) 

33. 

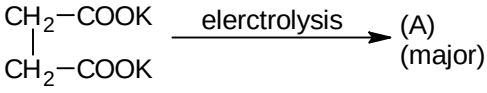


(X) & (Y) are

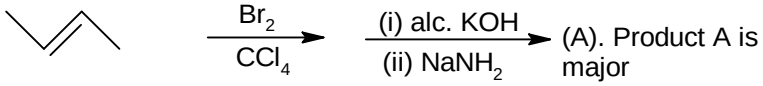
(A) Enantiomers
(B) Diastereomers
(C) E & Z isomer
(D) Constitutional isomer

34. The separation of racemic mixture into pure enantiomers is termed as
(A) Racemization
(B) Isomerization
(C) Resolution
(D) Equilibration

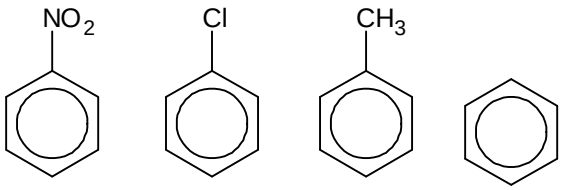
35. Which of the following compounds can be best prepared by Wurtz reaction?
(A) Isobutane
(B) n – butane
(C) n – pentane
(D) Iso – pentane

36. 

Product A is
(A) $CH_3-\!\!-\!\!-CH_3$
(B) $CH_2=\!\!=CH_2$
(C) $CH_3-\!\!-\!\!-CH=\!\!=CH_2$
(D) None

37. 

(A) $H_2C=CH-CH=CH_2$
(B) $CH_3-C\equiv C-CH_3$
(C) $CH_3-CH_2-C\equiv CH$
(D) $CH_3-CH=C=CH_2$

38. 

(A) (B) (C) (D)

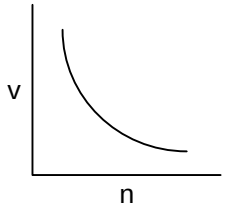
Correct order of rate of EAS (electrophilic aromatic substitution) is
(A) C > B > A > D
(B) C > D > A > B
(C) A > B > C > D
(D) C > D > B > A

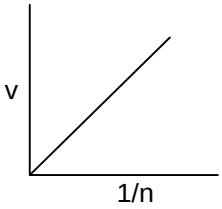
39. Which of the following is not a green house gas (GHG)?
(A) CO_2
(B) N_2O
(C) O_3
(D) NO

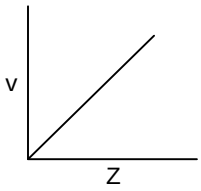
40. Rearrange the following (I to IV) in the order of increasing masses:
(I) 0.5 mole of O_3 (II) 0.5 gm atom of oxygen
(III) 3.011 x 10^{23} molecules of O_2 (IV) 5.6 litre of CO_2 at STP
(A) II < IV < III < I
(B) II < I < IV < III
(C) IV < II < III < I
(D) I < II < III < IV

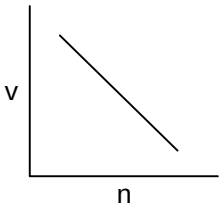
41. The number of neutrons in 5 g of D_2O (D is $^{2}_{1}H$) are
(A) 0.25 N_A
(B) 2.5 N_A
(C) 1.1 N_A
(D) None of these

42. Select the incorrect graph for velocity of e^- in an orbit vs. Z, $\frac{1}{n}$ and n

(A) 

(B) 

(C) 

(D) 

43. The orbital angular momentum of 3p electron is
(A) $\sqrt{3}$ h
(B) $\sqrt{6}$ h
(C) zero
(D) $\sqrt{2}\ \frac{h}{2\pi}$

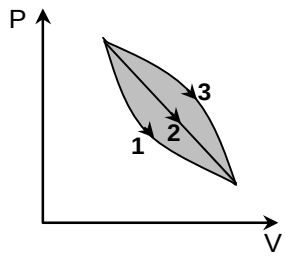
44. Air entering the lungs ends up in tiny sacs called alveoli. It is from the alveoli that oxygen diffuses into the blood. The average radius of the alveoli is 0.0050 cm and the air inside contains 14 per cent oxygen. Assuming that the pressure in the alveoli is 1.0 atm and the temperature is 37°C, calculate the number of oxygen molecules in one of the alveoli.
(A) 6×10^{13}
(B) 10^{24}
(C) 1.7×10^{22}
(D) 1.7×10^{12}

45. Starting out on a trip into the mountains, you inflate the tires on your automobile to a recommended pressure of 3.21×10^{5} Pa on a day when the temperature is –5.0°C. You drive to the beach, where temperature is 28°C. Assume that the volume increased by 3%. What is the final pressure in the tires?
(A) 350 Pa
(B) 3500 Pa

(C) 3.5×10^5 Pa
(D) None of these

46. A sample of liquid in a thermally insulated container (a calorimeter) is stirred for 2 hr. by a mechanical linkage to a motor in the surrounding, for this process:
(A) $w < 0;\ q = 0;\ \Delta U = 0$
(B) $w > 0;\ q > 0;\ \Delta U > 0$
(C) $w < 0;\ q > 0;\ \Delta U = 0$
(D) $w > 0;\ q = 0;\ \Delta U > 0$

47. A given mass of gas expands from the state A to the state B by three paths 1, 2 and 3 as shown in the figure. If w_1, w_2 and w_3 respectively be the work done by the gas along three paths then:



(A) $w_1 > w_2 > w_3$
(B) $w_1 < w_2 < w_3$
(C) $w_1 = w_2 = w_3$
(D) $w_2 < w_3 < w_1$

48. Which of the following relations is correct for k_f and k_b in an equilibrium process that contains equal moles of reactants and products?
(A) $k_f = k_b$
(B) $k_f > k_b$
(C) $k_f < k_b$
(D) We cannot predict

49. When Mg_2C_3 is reacted with water, a compound X is formed which on heating with Na followed by treatment with C_2H_5Br gives another compound Y. Calculate the number of sigma bond in Y.
(A) 10
(B) 6
(C) 8
(D) 12

50. Given the following K_a values, determine which species is the strongest base,
$HSO_4^- = 1.2\times10^{-2}, H_2PO_4^- = 6.3\times10^{-8}, HCO_3^- = 4.7\times10^{-11}$
(A) CO_3^{2-}
(B) H_2SO_4
(C) SO_4^{2-}
(D) HPO_4^{2-}

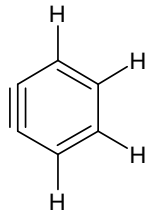
51. Given that K_w for water is 10^{-13} M^2 at 62^oC, compute the sum of pOH and pH for a neutral aqueous solution at 62^oC.
(A) 7.0
(B) 13.30
(C) 14.0
(D) 13.0

52. In the periodic table, metallic character of the elements shows which one of the following trends?
(A) Decreases down the group and increases across the period
(B) Increases down the group and decreases across the period
(C) Increase across the period and also down the group
(D) Decreases across the period and also down the group

53. Which of the following is arranged in decreasing order of size?
(A) $Mg^{2+} > Al^{3+} > O^{2-}$
(B) $O^{2-} > Mg^{2+} > Al^{3+}$
(C) $Al^{3+} > Mg^{2+} > O^{2-}$
(D) $Al^{3+} > O^{2-} > Mg^{2+}$

54. Correct statement regarding molecules SF_4, CF_4 and XeF_4 are
(A) 2, 0 and 1 lone pairs of central atom respectively
(B) 1, 0 and 1 lone pairs of central atom respectively
(C) 0, 0 and 2 lone pairs of central atom respectively
(D) 1, 0 and 2 lone pairs of central atom respectively

55. How many sp^2 and sp-hybridised carbon atoms are present respectively in the following compound?
(A) 4, 2
(B) 6, 0
(C) 3, 3
(D) 5, 1



56. For the alkali metals, which of the following increases with increasing atomic number?
(A) First ionization energy
(B) Electronegativity
(C) Hydration energy of the univalent ion
(D) Atomic radius

57. A solution of sodium in liquid ammonia is blue in colour due to:
(A) the presence of ions Na^+
(B) the presence of ammoniated electron
(C) the formation of $NaNH_2$
(D) the formation of sodium hydride

58. Colour of the bead in borax bead test is mainly due to the formation of
(A) metal oxides
(B) boron oxide
(C) metal metaborates
(D) elemental boron

59. PbI_4 does not exist because:
(A) iodine is not reactive
(B) Pb(IV) is oxidizing and I^- is strong reducing agent
(C) Pb(IV) is less stable than Pb (II)
(D) Pb^{4+} is not easily formed

60. The equilibrium constant K and reaction quotient Q are in ratio 0.33 : 1. It means that
(A) The reaction mixture will equilibrate to form more reactant species
(B) The reaction mixture will equilibrate to form more product species
(C) The equilibrium ratio of reactant to product concentrations will be 3
(D) The equilibrium ratio of reactant to product concentrations will be 0.33

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The coefficient of the term independent of x in the expansion of

$(1 + x + 2x^3)\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ is

(A) 1/3
(B) 19/54
(C) 17/54
(D) 1/4

62. The sum to (n + 1) terms of the series $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5}$ + ... is

(A) $\frac{1}{n+1}$
(B) $\frac{1}{n+2}$
(C) $\frac{1}{n(n+1)}$
(D) none of these

63. The number of integral values of λ for which $x^2 + y^2 + \lambda x + (1 - \lambda)y + 5 = 0$ is the equation of a circle whose radius cannot exceed 5, is
(A) 14
(B) 18
(C) 16
(D) None of these

64. The locus of the mid point of a chord of the circle $x^2 + y^2 = 4$ which subtends a right angle at the origin is
(A) $x + y = 2$
(B) $x^2 + y^2 = 1$
(C) $x^2 + y^2 = 2$
(D) $x + y = 1$

65. If α and β are the eccentric angles of the extremities of a focal chord of an ellipse, then the eccentricity of the ellipse is

(A) $\frac{\cos\alpha + \cos\beta}{\cos(\alpha - \beta)}$
(B) $\frac{\sin\alpha - \sin\beta}{\sin(\alpha - \beta)}$
(C) $\frac{\cos\alpha - \cos\beta}{\cos(\alpha - \beta)}$
(D) $\frac{\sin\alpha + \sin\beta}{\sin(\alpha + \beta)}$

66. The number of real tangents that can be drawn to the ellipse $3x^2 + 5y^2 = 32$ passing through (3, 5) is
(A) 0
(B) 1
(C) 2
(D) infinite

67. If F_1 and F_2 be the feet of the perpendicular from the foci S_1 and S_2 of an ellipse $\frac{x^2}{5} + \frac{y^2}{3} = 1$ on the tangent at any point P on the ellipse, then $(S_1 F_1)(S_2 F_2)$ is equal to
(A) 2
(B) 3
(C) 4
(D) 5

68. The normal to the rectangular hyperbola $xy = c^2$ at the point 't_1' meets the curve again at the point 't_2'. Then the value of $t_1^3 t_2$ is
(A) 1
(B) c
(C) –c
(D) –1

69. If PQ is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that OPQ is an equilateral triangle, O being the center of the hyperbola. then the 'e' eccentricity of the hyperbola, satisfies
(A) $1 < e < \frac{2}{\sqrt{3}}$
(B) $e = \frac{2}{\sqrt{3}}$
(C) $e = \frac{\sqrt{3}}{2}$
(D) $e > \frac{2}{\sqrt{3}}$

70. The slope of the tangent of the circle $(x - 6)^2 + y^2 = 2$, which is passing through the focus of the parabola $y^2 = 16x$ is
(A) ± 1
(B) ± 2
(C) –1/2, 2
(D) –2, 1/2

71. The condition that the parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ have a common normal other than x-axis (a, b, c being distinct positive real numbers) is
(A) $0 < \frac{b}{a-c} < 1$
(B) $\frac{b}{a-c} > 2$

(C) $\frac{b}{a-c} < 0$

(D) $1 < \frac{b}{a-c} < 2$

72. The number of ways in which 3 boys and 4 girls be seated around a circular table if no two boys sit together is
(A) 36
(B) 3!4!
(C) $7! - 3! \times 4!$
(D) 18

73. The number of positive integral solution of the equation xyz = 3000 are
(A) 200
(B) 300
(C) 3000
(D) 50

74. The sum of the divisors of $2^5.3^4.5^2$ is:
(A) $3^3.7^1.11^2$
(B) $3^2.7^1.11^2.31$
(C) 3.7.11.31
(D) none of these

75. The median AD of a triangle ABC is perpendicular to AB, Then value of tanA + 2tanB is
(A) 1
(B) 0
(C) 2
(D) 3

76. If in a triangle ABC, cosA cosB + sinA sinB sinC = 1, then $a : b : c =$
(A) $1:1:\sqrt{2}$
(B) 1:2:3
(C) 1:3: 4
(D) None of these

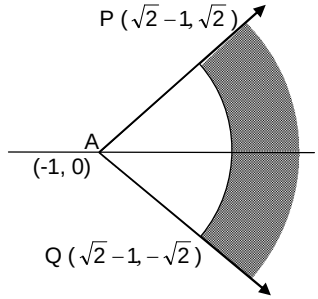
77. The value of $\cos\frac{\pi}{7} \cos \frac{2\pi}{7} \cos\frac{3\pi}{7}$ is
(A) 1/8
(B) –1/8
(C) 1
(D) 0

78. The minimum value of $\cos 2\theta + \cos \theta$ for real values of θ is
(A) –9/8
(B) 0
(C) –2
(D) none of these

79. For all complex numbers z_1, z_2 satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$, the minimum value of $|z_1 - z_2|$ is
(A) 0
(B) 2
(C) 7
(D) 17

80. If the line $y = \sqrt{3}x$ cuts the curve $x^3 + 3y^2 + 4x + 5y - 1 = 0$ at the points A, B, C then $OA \cdot OB \cdot OC$ is
(A) $\frac{4}{13}\left(3\sqrt{3} - 1\right)$
(B) $3\sqrt{3} + 1$
(C) $\frac{2}{\sqrt{3}} + 7$
(D) None of these

81. The locus of z which lies in shaded region is best represented by
(A) $z : |z+1| > 2, |arg(z+1)| < \pi/4$
(B) $z : |z-1| > 2, |arg(z-1)| < \pi/4$
(C) $z : |z+1| < 2, |arg(z+1)| < \pi/2$
(D) $z : |z-1| < 2, |arg(z-1)| < \pi/2$



82. If (b+c),(c+a),(a+b) are in H.P, then a^2, b^2, c^2 are in……
(A) A.P.
(B) G.P.
(C) H.P.
(D) None

83. Total number of integral solution of the equation xyz=24 is
(A) 30
(B) 120
(C) 85
(D) 60

84. The coefficient of the term independent of x in the expansion of $\left(\sqrt{\frac{x}{3}} + \frac{3}{2x^2}\right)^{10}$ is
(A) 5/4
(B) 7/4
(C) 9/4
(D) none of these

85. If a, b, c are in H.P., then the value of $\left(\frac{1}{b} + \frac{1}{c} - \frac{1}{a}\right)\left(\frac{1}{c} + \frac{1}{a} - \frac{1}{b}\right)$ is
(A) $\frac{2}{bc} - \frac{1}{b^2}$

(B) $\frac{1}{5}\left(\frac{3}{c^2}+\frac{2}{ca}-\frac{1}{a^2}\right)$

(C) $\frac{6}{b^2}-\frac{1}{ab}$

(D) none of these

86. The sum to n terms of the series $1^2 + (1^2 + 3^2) + (1^2 + 3^2 + 5^2) + \ldots$ is

(A) $\frac{1}{6}n(n+1)(2n^2+2n-1)$

(B) $\frac{1}{3}\left(n^4+2n^2\right)$

(C) $\frac{1}{3}\left(n^3+3n^2-n\right)$

(D) none of these

87. If $a_1, a_2, \ldots, a_n$ are positive real numbers whose product is a fixed number c, then the minimum value of $a_1 + a_2 + \ldots + a_{n-1} + 2a_n$ is

(A) $n(2c)^{1/n}$

(B) $(n + 1)c^{1/n}$

(C) $2nc^{1/n}$

(D) $(n + 1)(2c)^{1/n}$

88. The integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots are

(A) (1, 6)

(B) (8, 12)

(C) (4, 6)

(D) (3, 15)

89. If $\sin\alpha, \cos\alpha$ are the roots of the equation $ax^2 + bx + c = 0$, $(c \neq 0)$ then

(A) $a^2 - b^2 + 2ac = 0$

(B) $(a + c)^2 = b^2 - c^2$

(C) $a^2 + b^2 - 2ac = 0$

(D) $(a - c)^2 = b^2 + c^2$

90. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$, $(abc \neq 0)$ is equal to sum of the squares of their reciprocals, then a/c, b/a, c/b, are in

(A) arithmetic progression (A.P.)

(B) geometrical progression (G.P.)

(C) harmonic progression (H.P.)

(D) none of these